

Problem

- No lab access — visual inspection misses early contamination; lab tests are slow and costly.
- Sudden rain re-wets drying grain, driving mould growth, with no local warning.
- Scans go untracked, leaving no record for future decisions or buyers.

Project Objectives

- First-line risk check on any smartphone — no lab, no training.
- Two scan tiers: kernel image + chemical test strip.
- Every result paired with UNBS/MAAIF guidance.
- Rain and weather alerts protect drying grain.
- Scan history and exportable PDF reports.
- Voice assistant + Luganda for hands-busy, non-literate users.

Project Requirements

- R1** Secure auth — email, Google, OTP, biometric
- R2** On-device kernel classifier (TFLite, YOLOv8n-cls)
- R3** Test-strip optical-density + ppb reading
- R4** Standards-based guidance on every result
- R5** Daily + urgent rain alerts, even app closed
- R6** Hands-free voice assistant
- R7** Scan history + exportable PDF reports
- R8** Firestore storage, English/Luganda

Target Users

FARMERS

Scan at harvest, get instant results and rain alerts.

GRAIN HANDLERS & TRADERS

Screen batches before purchase, with traceable history.

EXTENSION OFFICERS & BUYERS

Get shared PDF reports and voice-summarised results.

Future Work

- Calibrate the strip reader against certified aflatoxin standards.
- Restrict strip scanning to in-app capture for reliability.
- Bring background rain alerts to iOS.
- Add a lab-referral pathway for high-risk results.

Conclusion

AflAlert turns a costly, inaccessible lab test into an everyday smartphone check — combining on-device classification, strip reading, rain alerts, and voice support to protect both harvest income and health.

References

[1] Flutter — flutter.dev/docs
[2] Firebase — firebase.google.com/docs
[3] TensorFlow Lite — tensorflow.org/lite
[4] Codex Alimentarius — CODEX STAN 193-1995
[5] UNBS — Maize grains specification
[6] MAAIF — Post-harvest handling guidelines
[7] flutter_tts & workmanager — pub.dev

Project Design — System Architecture

AflAlert — System Architecture

Presentation Layer — Flutter UI (lib/screens)

- Onboarding & Auth: welcome, login, registration, OTP
- Home: scan launcher, weather, alerts
- Kernel / Strip Scan: camera capture + results
- History & Reports: scan list, PDF export
- Notifications: alerts & updates
- Settings, Profile & Voice Assistant

Services Layer — Business Logic (lib/services)

- Kernel Classification: tflite_service (TFLite)
- Strip Analysis: line detection, OD → ppb
- Weather / Rain / Heat Alerts: morning_alert, rain_alert, temp_humidity_alert
- Voice Assistant: speech-to-text, TTS
- PDF & Report Storage: pdf_service, report_storage
- Firebase Access: auth, firestore, storage

Platform & External Services

- TensorFlow Lite: on-device ML model
- Firebase: Auth · Firestore · Storage · Cloud Functions · App Check
- Android WorkManager: background alert scheduling
- Weather API: forecast data
- Device Camera: kernel & strip photo capture

Screens call into services; services own all business logic and talk to Firebase, the bundled TFLite model, the device camera, and platform schedulers. This keeps each detection tier (kernel vs. strip) and supporting feature independently testable.

Flutter app (Android/iOS/Web) with a services layer for classification, strip analysis, alerts, voice, and reports — backed by Firebase (Auth, Firestore, Storage, Cloud Functions). Kernel photos run on-device via TFLite; strips are read with deterministic optical-density processing, no model required.

4,892

labelled kernel images

98.8%

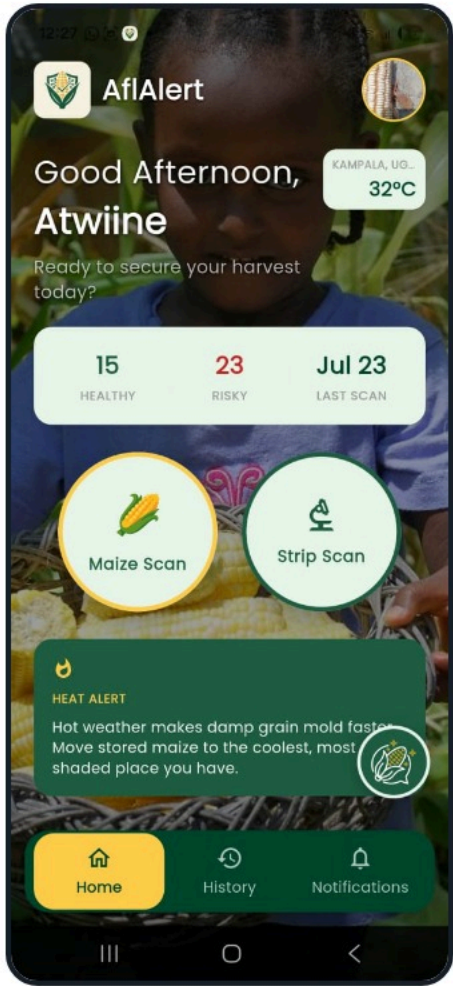
top-1 validation accuracy

~0.5ms

per image, on-device

RESULTS

Step 1 — Key Components

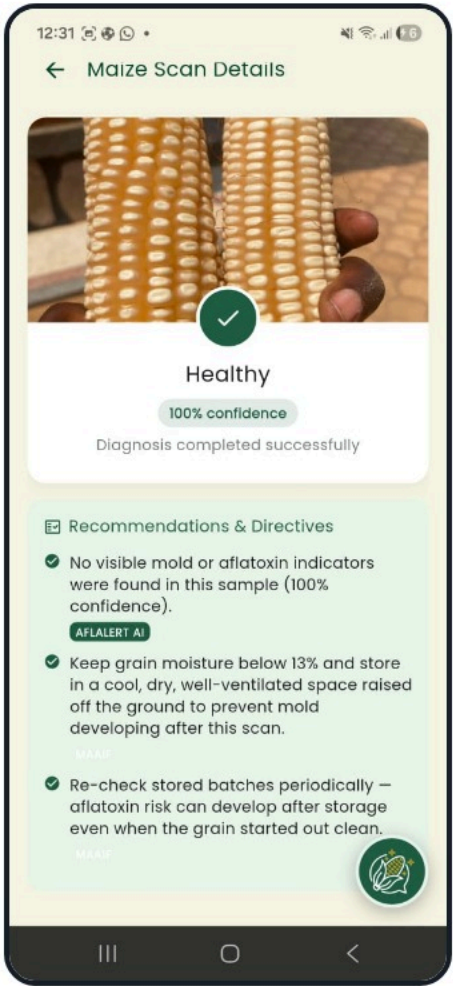


Maize Scan
On-device TFLite classifier

Strip Scan
Optical-density reader

One home screen, two scan tiers, proactive weather warnings.

Step 2 — Verdict: Healthy

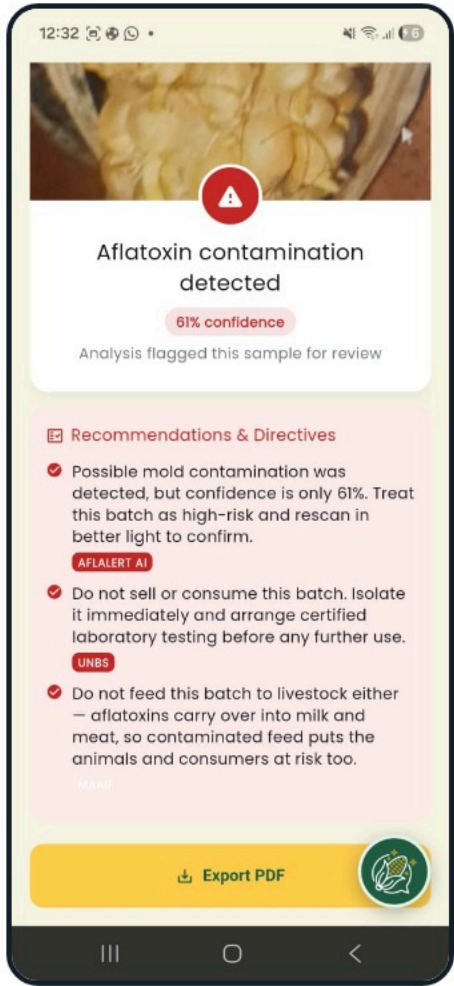


Confidence
100% softmax score

Guidance
MAAIF storage tips

Runs fully on-device; even safe results ship storage guidance.

Step 3 — Risk Detected

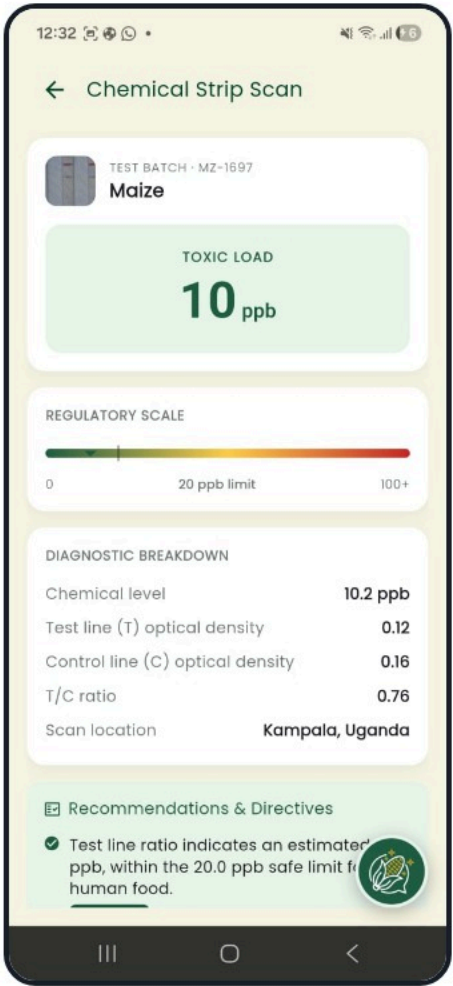


Flagged
61% confidence, shown not hidden

Referral
Isolate batch, confirm in lab

Above-threshold results trigger an isolate-and-confirm directive.

Step 4 — Strip Reading, ppb



Toxic load
10 ppb

Reg. scale
vs 20 ppb limit

Quantitative ppb reading from a lateral-flow test strip.

Repository: github.com/priscillanabachwa/AFLALERT

Live app: aflalert.web.app

Project site: naluyimalaureen-source.github.io/AFLALERT-website